Skill Issue or Scoring Issue?

A Probability Study of Pickleball vs Tennis Scoring

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Introduction

In many competitive games, the better player does not always win. This raises a natural probabilistic question: how much does the *scoring system* influence match outcomes?

Even when one player has a small advantage in individual rallies, scoring rules may amplify or weaken that advantage. This project studies how scoring systems convert rally-level skill into overall match results.

This project compares **pickleball** and **tennis**. Both sports involve rally-based gameplay, but their scoring systems differ. In tennis, every rally produces a point. In pickleball, only the serving player can score (side-out scoring). As a result, winning rallies does not always translate directly into points.

The central question is:

If one player is slightly better at winning rallies, which scoring system is more likely to allow that player to win the match?

The analysis focuses on:

- probability that the stronger player wins
- upset probability
- comeback probability from disadvantaged states

All estimates use sampling and bootstrap methods from CS109.

Probabilistic Model

Each rally is modeled as a random variable. Let

$$X_i = \begin{cases} 1 & \text{if Player A wins rally } i \\ 0 & \text{otherwise} \end{cases}$$

and assume

$$X_i \sim \text{Bernoulli}(p),$$

where $p = P(\text{A wins a rally})$. When $p = 0.5$, both players have equal skill; when $p > 0.5$, Player A is stronger. Rallies are assumed independent.

For n rallies, the number won by Player A is

$$S_n = \sum_{i=1}^{n} X_i.$$

Since each X_i is Bernoulli,

$$S_n \sim \text{Binomial}(n, p),$$

with

$$E[X_i] = p, \qquad \mathrm{Var}(X_i) = p(1-p),$$

and

$$E[S_n] = np, \qquad \mathrm{Var}(S_n) = np(1-p).$$

These formulas describe the expected rally advantage and its variability. The scoring system then determines how this rally-level advantage translates into points and match outcomes.

Scoring Systems

Pickleball. Games are played to 11 points and must be won by two. Only the serving player can score. If the receiver wins a rally, the serve changes but no point is awarded.

Tennis. A set consists of games, each won by reaching four points with a margin of two. A set is won by reaching six games with a margin of two. Every rally contributes directly to the score.

Because pickleball requires the serve to score, rally outcomes and score progression are partially decoupled.

Sampling and Estimation

Exact match probabilities are difficult to compute analytically because many score states are possible. Instead, these probabilities are estimated using repeated sampling.

For a fixed value of p, $N = 2000$ independent matches are simulated using the Bernoulli rally model. Let

$$W_j = \begin{cases} 1 & \text{if Player A wins match } j \\ 0 & \text{otherwise} \end{cases}$$

be the indicator variable for the outcome of simulated match j.

The estimated win probability is the sample mean

$$\hat{p} = \frac{1}{N} \sum_{j=1}^{N} W_j.$$

Since each W_j is Bernoulli, $\hat{p}$ is the mean of $N = 2000$ independent Bernoulli trials. By the **Central Limit Theorem**, for large N the sampling distribution of $\hat{p}$ is approximately normal:

$$\hat{p} \approx \mathcal{N}\left(p, \frac{p(1-p)}{N}\right).$$

This leads to the standard normal approximation for the standard error,

$$\mathrm{SE}(\hat{p}) \approx \sqrt{\frac{p(1-p)}{N}},$$

which is used to construct the confidence intervals around $\hat{p}$.

Bootstrap uncertainty. In addition to the CLT approximation, uncertainty is estimated using bootstrap resampling. For each setting, $B = 1000$ bootstrap samples are generated by sampling with replacement from the simulated match outcomes and recomputing the sample mean. The resulting bootstrap distribution provides standard errors and confidence intervals.

Results

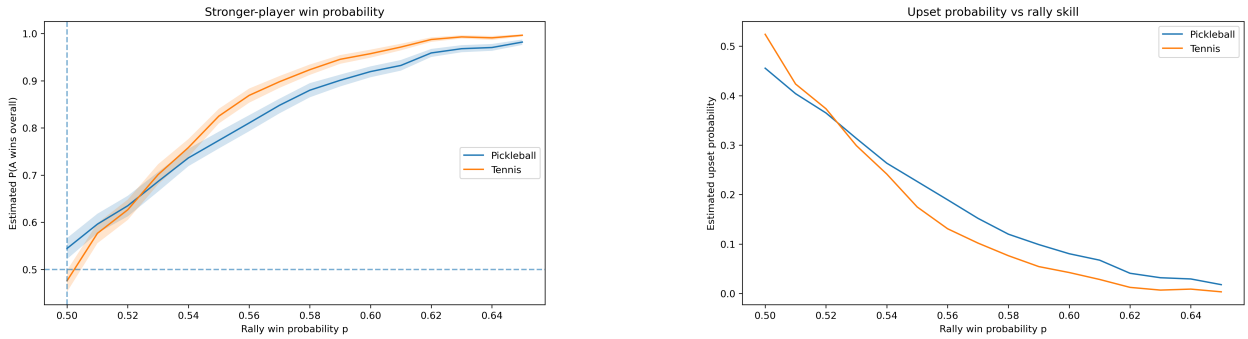

Figure 1: Left: stronger-player win probability vs. rally skill p. Right: upset probability vs. rally skill.

The probability that the stronger player wins increases as p increases, but the increase is steeper for tennis than for pickleball. This means tennis scoring converts small rally-level skill advantages into overall wins more efficiently.

Equivalently, upset probability is consistently higher under pickleball scoring. The gap between the two sports is much larger than the estimated uncertainty, and the bootstrap intervals agree closely with the CLT normal approximation.

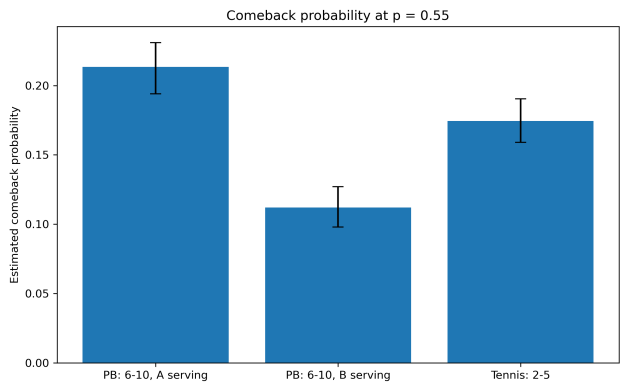

Figure 2: Estimated comeback probabilities from disadvantaged score states. Error bars show bootstrap confidence intervals.

Comeback probability in pickleball depends strongly on serving status. When trailing 6–10, Player A has a much higher chance of winning when serving than when receiving. This happens because only the server can score points. In tennis, every rally produces a point, so comeback probability depends less on serve transitions.

Conclusion

Scoring systems significantly affect how skill differences translate into match outcomes. Tennis scoring allows rally advantages to accumulate directly into points, making the stronger player more likely to win.

In contrast, pickleball's side-out scoring introduces additional randomness by separating rally outcomes from scoring opportunities. This leads to higher upset probabilities and comeback dynamics that depend strongly on serving.

More broadly, this project illustrates a central idea from CS109: probabilistic behavior depends not only on the underlying random variables, but also on the structure of the system that aggregates them.